TRI-OCTAGON MATHEMATICAL RESEARCH

Native Potential Descent and Closed Coherence Coordinates in the Tri-Octagon Map

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Abstract

The native Tri-Octagon map combines a quartic-potential gradient prestage with simultaneous harmonic-three phase synchronization. We establish two distinct descent statements for its existing potential. On the admitted positive in-phase triad branch with $g > 0$, strict normal stability implies strict local descent modulo common phase. The proof identifies the actual potential Hessian and shows that the triad quotient response contracts this form, even when positive diagonal balance fails. Separately, at a fixed rational coefficient tuple, an inherited invariant domain with amplitudes between 0.9 and 1.1 and ordinary phase width at most $\pi/12$ admits the explicit decrease bound 16869/20000 times the squared prestage increment. Equality in monotonicity occurs exactly on the unique aligned fixed circle. On that same finite domain, the three intensities and two signed pair areas reconstruct full complex coherence, define an autonomous map on their exact image, and retain every matched edge energy, the potential, and the descent quantity. Two additional real pair entries are minimal within the prescribed raw-entry dictionary; restricting the dictionary to real coherences requires three. Static same-data families prove these lower bounds by failure of current energy retention. The results concern the ideal mathematical recurrence and do not identify the potential with physical energy or reconstruct the discarded common-phase history.

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1 Question and contributions

The native recurrence already selects a quartic onsite potential and a quadratic graph term. Its first stage is a unit negative-gradient increment of that potential. Its second stage rotates the three components by different phase kicks. Because these rotations preserve intensities but can change graph distances, the gradient identity alone does not prove descent of the complete map.

The first question is therefore about a specified scalar, not the existence of some auxiliary Lyapunov function: when does the complete update decrease the native potential? The second asks how much of the existing pair information can be retained while preserving both autonomous evolution and that energy accounting.

The response, transport, and closure results of [1] and the coherent-domain results of [2] are inherited inputs. They establish the admitted positive branch, its exact derivative and triad spectrum, regular coherence closure, an invariant sector, and a unique aligned profile. The present paper assembles the accepted deductions of two subsequent investigations [3, 4]:

1. strict nonlinear local descent in the potential's own Hessian, together with its coexistence with failed positive diagonal balance;
2. an explicit finite-domain descent inequality and complete monotonicity-equality classification;
3. five real closed coordinates preserving all required energy quantities, with attained minimum counts in two fixed raw-entry dictionaries.

The proofs are included here. Inherited convergence is not used as a substitute for a monotonicity argument, and a current-energy obstruction is not described as dynamic nonclosure.

2 Native map, scalar, and distinct settings

Let $\Omega \in \mathbb{C}^3$, let the coefficients be real, and set

$$L = \mathbf{1}\mathbf{1}^T - 3I = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}. \tag{2.1}$$

The prestage v, increment b, and complete map are

$$v = \Omega + \varepsilon\Omega \odot (k - |\Omega|^2) + gL\Omega, \qquad b = v - \Omega, \tag{2.2}$$

$$\psi_i = \mathrm{Arg}_0(v_i), \qquad \delta_i = \lambda \sum_{j \neq i} \sin 3(\psi_j - \psi_i),$$

$$F_i(\Omega) = v_i e^{i\delta_i}. \tag{2.3}$$

All kicks use the same prestage. The native convention is $\mathrm{Arg}_0(0) = 0$; at $\lambda = 0$, the implemented synchronizer returns the prestage directly. No normalization, added clock, or modified zero convention is introduced.

The scalar throughout is

$$\mathcal{V}(\Omega) = \varepsilon \sum_i \left(\frac{|\Omega_i|^4}{4} - \frac{k_i|\Omega_i|^2}{2} \right) + \frac{g}{2} \sum_{i<j} |\Omega_i - \Omega_j|^2. \tag{2.4}$$

Each undirected edge appears once. In nodewise real coordinates $x_i \in \mathbb{R}^2$,

$$\nabla_{x_i}\mathcal{V} = \varepsilon(|x_i|^2 - k_i)x_i + g\sum_{j \neq i}(x_i - x_j), \qquad b = -\nabla_{\mathbb{R}^6}\mathcal{V}. \tag{2.5}$$

This is the inherited normalization of [1], response chapter, `eq:potential`. The separate harmonic-three phase potential is a different function; descent of that function does not establish descent of (2.4).

Whenever all prestage components are nonzero, the sine in (2.3) equals the imaginary part of

$$\left(\frac{v_j\overline{v}_i}{|v_j||v_i|}\right)^3.$$

The complete map is therefore smooth and common-phase equivariant on that regular domain without reliance on a principal-argument cut. The potential is common-phase invariant. Unrestricted quotient assertions at zero prestages would conflict with the native convention; the accepted counterexample in [1], `eq:zerocounter`, remains in force.

Table 1: The settings are separate. The finite-domain result does not require identifying that domain with an unspecified local neighborhood.

Result	Hypotheses	Conclusion
Local descent	Admitted positive in-phase triad branch; $g > 0$; all normal multipliers strictly inside the unit disk	Descent in some regular neighborhood of the fixed circle
Finite-domain descent	Exact tuple (4.1); closed domain (4.2)	Explicit one-step inequality and equality circle
Closed coordinates and minimum	The same tuple and domain; all intensities always retained	Invariant retained image, exact energy retention, restricted raw-entry minima

The response matrix H_r and radial derivative A_r below are not the coherence matrix $\mathcal{H} = \Omega\Omega^\dagger$ or the signed areas A_{ij}. The prestage v is likewise distinct from $\mathcal{V}$.

3 Local descent and diagonal balance

The positive branch and its complete derivative are inherited from [1], response chapter, `thm:branch`, `eq:weighted`, and `eq:SP`. Start from a strictly normally stable equal triad and continue the positive in-phase branch only within its admitted neighborhood. At a chosen profile $r > 0$,

$$\varepsilon r_i(k_i - r_i^2) + g(Lr)_i = 0. \tag{3.1}$$

Write

$$\begin{aligned} R &= \operatorname{diag}(r), & H_r &= I + \varepsilon\operatorname{diag}(k - r^2) + gL, & H_r r &= r, \\ P_{\mathrm{resp}} &= R^{-1}H_r R, & A_r &= H_r - 2\varepsilon R^2, & & \\ J &= (I + 3\lambda L)P_{\mathrm{resp}}. & & & & \end{aligned} \tag{3.2}$$

The order in J is synchronization after prestage. The common phase satisfies $P_{\mathrm{resp}}\mathbf{1} = J\mathbf{1} = \mathbf{1}$.

Theorem 3.1 (Strict local native-potential descent)**.** *Fix an admitted positive profile (3.1) with $g > 0$. Assume every eigenvalue α_j of A_r has $|\alpha_j| < 1$, and both quotient phase multipliers $\rho_\pm$ defined in (3.11) have $|\rho_\pm| < 1$. There is a regular open neighborhood U of $\mathcal{C}_r = \{e^{i\theta}r : \theta \in \mathbb{R}\}$ such that*

$$\mathcal{V}(F\Omega) < \mathcal{V}(\Omega) \quad (\Omega \in U \setminus \mathcal{C}_r).$$

Every point of $\mathcal{C}_r$ is fixed. No explicit radius or maximal basin is asserted.

3.1 The actual potential Hessian

In the polar chart $\Omega = e^{i\theta}(r+a) \odot e^{i\phi}$, put $q = r + a > 0$. Direct substitution gives

$$\begin{aligned}\mathcal{V} = \varepsilon \sum_i \left(\frac{q_i^4}{4} - \frac{k_i q_i^2}{2} \right) + \frac{g}{2} \sum_{i<j} (q_i - q_j)^2 \\ + g \sum_{i<j} q_i q_j [1 - \cos(\phi_i - \phi_j)].\end{aligned} \tag{3.3}$$

The radial first variation vanishes by (3.1). Its Hessian is

$$B_a = \varepsilon \operatorname{diag}(3r^2 - k) - gL = I - A_r. \tag{3.4}$$

The phase term begins at degree two. Varying an amplitude in its coefficient contributes degree at least three; equivalently, its mixed second derivatives contain a sine vanishing at alignment. Thus there is no mixed quadratic amplitude–phase term.

The phase Hessian has off-diagonal entries $-g r_i r_j$ and diagonal entries $g r_i \sum_{j \neq i} r_j$. Using (3.1),

$$\begin{aligned} K_\phi = R(I - H_r)R = R^2(I - P_{\text{resp}}), \qquad & K_\phi \mathbf{1} = 0, \\ \phi^T K_\phi \phi = g \sum_{i<j} r_i r_j (\phi_i - \phi_j)^2. & \end{aligned} \tag{3.5}$$

Since $g > 0$, this form is positive on $\mathbb{R}^3/\operatorname{span}\{\mathbf{1}\}$ and has exactly the common-phase kernel. In any fixed local quotient section,

$$\mathcal{V}(\Omega) - \mathcal{V}(r) = \frac{1}{2} a^T B_a a + \frac{1}{2} \phi^T K_\phi \phi + O(\|(a, [\phi])\|^3). \tag{3.6}$$

Both factors $1/2$ are essential. Symmetry of A_r and $|\alpha_j| < 1$ make B_a positive definite.

3.2 Ordered response in the potential form

Differentiating the prestage at r gives $\delta v = A_r a + i H_r R \phi$. Its phase derivative is P_{resp}; differentiating the simultaneous kicks adds $3\lambda L$. Since synchronization preserves moduli, the complete derivative is $\operatorname{diag}(A_r, J)$ in amplitude/phase coordinates, with no first-order mixing.

Set $d = 1 - 9\lambda$. The triad identity for L gives

$$J = d P_{\text{resp}} + 3\lambda \mathbf{1}(\mathbf{1}^T P_{\text{resp}}). \tag{3.7}$$

The inherited weighted balance $P_{\text{resp}}^T R^2 = R^2 P_{\text{resp}}$ implies

$$P_{\text{resp}}^T K_\phi = R^2 P_{\text{resp}}(I - P_{\text{resp}}) = K_\phi P_{\text{resp}}. \tag{3.8}$$

Since $K_\phi \mathbf{1} = 0$, all common-phase terms disappear from the quadratic form:

$$J^T K_\phi J = d^2 P_{\text{resp}}^T K_\phi P_{\text{resp}}. \tag{3.9}$$

These are full-space identities; strict positivity or negativity is interpreted on the quotient. The Euclidean sum-zero plane is not assumed invariant under J.

For the exact inherited triad reduction [1], spectral chapter, `thm:triad`, let $\sigma = \sum_i r_i$ and

$$M_3 = \sigma R^{-1} - \mathbf{1}\mathbf{1}^T, \qquad H_r = I - g M_3. \tag{3.10}$$

The second identity follows from the fixed equations. Weighted Cauchy–Schwarz gives

$$z^T M_3 z = \sigma \sum_i z_i^2/r_i - (\sum_i z_i)^2 \geq 0,$$

with equality exactly when $z \parallel r$. Its other eigenvalues $\mu_\pm$ are positive. Choose Euclidean-orthonormal eigenvectors $w_\pm \perp r$, and put

$$f_\pm = R^{-1} w_\pm, \qquad p_\pm = 1 - g\mu_\pm, \qquad \rho_\pm = dp_\pm. \tag{3.11}$$

Then $P_{\mathrm{resp}} f_\pm = p_\pm f_\pm$, and (3.7) makes their classes eigenvectors of J on the quotient. Moreover,

$$f_\beta^T K_\phi f_\alpha = g\, w_\beta^T M_3 w_\alpha = g\mu_\alpha \delta_{\alpha\beta}. \tag{3.12}$$

Writing $[\phi] = c_+[f_+] + c_-[f_-]$, we obtain the exact decrement

$$\boxed{\phi^T (J^T K_\phi J - K_\phi)\phi = g \sum_{\alpha=\pm} \mu_\alpha(\rho_\alpha^2 - 1)c_\alpha^2.} \tag{3.13}$$

It is strictly negative on every nonzero quotient direction under the theorem's assumptions. Repeated positive eigenvalues at equal radii allow any orthonormal basis of their eigenspace; $d = 0$ also causes no difficulty.

3.3 Nonlinear completion

Proof of Theorem 3.1. In a Euclidean-orthonormal eigenbasis of the symmetric A_r, write a with coefficients t_j. Since $B_a = I - A_r$,

$$a^T (A_r^T B_a A_r - B_a) a = \sum_{j=1}^{3} (1 - \alpha_j)(\alpha_j^2 - 1) t_j^2. \tag{3.14}$$

Every coefficient is negative when $|\alpha_j| < 1$. This uses the actual potential Hessian, not an arbitrary energy form.

A legitimate quotient section is $E = \mathrm{span}\{f_+, f_-\}$: it complements $\mathbf{1}$, since $\mathbf{1}^T R^2 f_\pm = r^T w_\pm = 0$. Choose $\phi = c_+ f_+ + c_- f_-$, and after the complete update remove its local common phase to return to this section. Possible common-phase components of $J f_\pm$ are thereby removed; invariance of E under J is unnecessary.

For $u = (t_1, t_2, t_3, c_+, c_-)$, the induced smooth map is

$$u^+ = Tu + O(\|u\|^2), \qquad T = \mathrm{diag}(\alpha_1, \alpha_2, \alpha_3, \rho_+, \rho_-).$$

At the profile, input, prestage, and output moduli are $r_i > 0$. A sufficiently small chart preserves this regularity and the smooth output section. Substituting into (3.6) yields

$$\mathcal{V}(F\Omega) - \mathcal{V}(\Omega) = -\frac{1}{2}\left[\sum_j (1 - \alpha_j)(1 - \alpha_j^2) t_j^2 + g \sum_{\alpha=\pm} \mu_\alpha (1 - \rho_\alpha^2) c_\alpha^2\right] + O(\|u\|^3). \tag{3.15}$$

Let $m > 0$ be the minimum of the five bracket coefficients. Bound the absolute remainder by $C\|u\|^3$ on a smaller chart and shrink until $C\|u\| \leq m/4$. Then the change is at most $-m\|u\|^2/4$, strictly negative unless $u = 0$. The coordinate norm only bounds the remainder; the decreased scalar is still $\mathcal{V}$.

Common-phase equivariance of the regular map and invariance of $\mathcal{V}$ rotate this neighborhood around the entire fixed circle with the same conclusion. A sufficiently small rotation-saturated open neighborhood is regular. Its equality states have $u = 0$, exactly $\mathcal{C}_r$. □

Unit-modulus normal multipliers are excluded. A semidefinite quadratic decrement alone would not justify this nonlinear conclusion.

3.4 Coexistence with failed diagonal balance

The inherited balance criterion asks for a positive *diagonal* W satisfying $WJ = J^TW$. Near a strictly stable equal triad, all off-diagonal entries of J are positive. At equality they are $q_0 = g + \ell - 3g\ell$, where $\ell = 3\lambda$; the transverse multiplier $1 - 3q_0 \in (-1, 1)$ gives $0 < q_0 < 2/3$, and positivity persists nearby.

In this neighborhood the criterion is exactly

$$\mathcal{C}_{\mathrm{cyc}} = J_{12}J_{23}J_{31} - J_{21}J_{32}J_{13} = 0. \tag{3.16}$$

Necessity follows by multiplying the three pair-balance relations. For sufficiency the positive weights $(J_{21}J_{31}, J_{12}J_{31}, J_{13}J_{21})$ balance two edges identically and leave the cycle difference on the third [1], `thm:cycle`.

For $\sigma_1 = r_1+r_2+r_3$, $\sigma_2 = r_1r_2+r_1r_3+r_2r_3$, $\sigma_3 = r_1r_2r_3$, and $d = 1-3\ell$, the source factorization is

$$\begin{aligned} \mathcal{C}_{\mathrm{cyc}} &= \frac{g\ell d(r_1 - r_2)(r_1 - r_3)(r_2 - r_3)}{\sigma_3^2}\,\mathcal{K}, \\ \mathcal{K} &= g^2\ell\sigma_1^3 + [g^2(1 - 2\ell) - g\ell]\sigma_1\sigma_2 + (3g\ell - g + \ell)\sigma_3. \end{aligned} \tag{3.17}$$

The scalar $\mathcal{K}$ is not the phase form K_ϕ.

Along the inherited weak branch $k_i = k + \eta\kappa_i$, with $k > 0$ and $\sum_i \kappa_i = 0$,

$$r_i = \sqrt{k} + \eta\frac{\varepsilon\sqrt{k}}{2\varepsilon k + 3g}\kappa_i + O(\eta^2).$$

Substitution in (3.17) gives

$$\begin{aligned} \mathcal{C}_{\mathrm{cyc}} &= C_3\eta^3 + O(\eta^4), \\ C_3 &= g\ell d K_0 \left(\frac{\varepsilon}{2\varepsilon k + 3g}\right)^3 \prod_{i<j}(\kappa_i - \kappa_j), \qquad K_0 = g(9g - 1) + \ell(1 - 3g)^2. \end{aligned} \tag{3.18}$$

If $C_3 \neq 0$, all sufficiently small nonzero η have nonzero cycle difference while branch positivity and strict normal stability persist. Theorem 3.1 therefore gives local potential descent despite failed diagonal balance. An unspecified sufficiently small interval does not certify any particular numerical heterogeneity.

This comparison separates four properties. Diagonal balance is a nodewise weighting condition. General SPD self-adjointness allows a dense positive metric and is equivalent to a real semisimple full spectrum [1], `thm:spd`. In the stable triad, the real semisimple quotient spectrum is separated from the common eigenvalue 1, so such a full SPD symmetrizer exists. Strict normal spectral stability bounds multiplier moduli. Contraction of this specified potential requires the additional identities (3.4) and (3.13), followed by the remainder argument. These claims are not interchangeable.

Indeed $J^TK_\phi = K_\phi J = dK_\phi P_{\mathrm{resp}}$, but K_ϕ is singular on the full phase space and positive only on the quotient. It is not a full-space SPD metric, and no independently selected metric replaces the native potential in the proof.

4 Explicit descent on the coherent domain

We now fix a parameter tuple and a finite domain independently of the preceding local continuation hypotheses:

$$\varepsilon = \frac{1}{10}, \qquad g = \frac{1}{60}, \qquad \lambda = \frac{1}{300}, \qquad k = \left(\frac{999}{1000}, 1, \frac{1001}{1000}\right). \tag{4.1}$$

With $D_0 = \pi/12$, let

$$\mathcal{D} = \left\{\Omega \in \mathbb{C}^3 : \frac{9}{10} \le |\Omega_i| \le \frac{11}{10},\ \text{ordinary phase width} \le D_0\right\}. \tag{4.2}$$

The containing phase interval can have any common orientation. It is an ordinary full-turn interval, not phase modulo $2\pi/3$.

We use the following precise inherited facts from [2], Sections 6–7, `lem:phase`, `eq:lower`, `eq:upper`, and `thm:convergence`:

1. Positive prestage mixing keeps its phases in the input interval, and

$$\frac{88701}{100000} + \frac{3}{100}\cos D_0 \le |v_i| \le \frac{107701}{100000}, \qquad \frac{9}{10} < |v_i| < \frac{11}{10}. \tag{4.3}$$

2. For positive-modulus prestages of ordinary phase width $D_v < \pi/6$ and $0 \le \lambda \le 1/9$, the simultaneous phase stage preserves its interval and satisfies

$$|\psi_i^+ - \psi_j^+| \le (1 - 9\lambda\cos 3D_v)|\psi_i - \psi_j|. \tag{4.4}$$

3. The domain $\mathcal{D}$ is forward invariant. There is a unique aligned fixed amplitude profile $r_* \in [9/10, 11/10]^3$, satisfying (3.1) at (4.1). Its common rotations are fixed states.

These inputs include positive *prestage* bounds; positive input amplitudes alone would not justify the regularity needed later. No profile value is numerically computed.

Theorem 4.1 (Finite-domain native descent)**.** *For every $\Omega \in \mathcal{D}$ at (4.1),*

$$\boxed{\mathcal{V}(F\Omega) - \mathcal{V}(\Omega) \le -\frac{16869}{20000}\|v - \Omega\|_2^2.} \tag{4.5}$$

Equality in monotonicity, $\mathcal{V}(F\Omega) = \mathcal{V}(\Omega)$, holds exactly on $\{e^{i\theta}r_* : \theta \in \mathbb{R}\}$. The coefficient is a valid bound, without an optimality claim, and multiplies the squared prestage increment.

4.1 A whole-segment curvature estimate

Proof. Both Ω and v lie in the convex product of closed radius-11/10 disks. For $b = v - \Omega$,

$$|\Omega_i + tb_i| \le (1-t)|\Omega_i| + t|v_i| \le 11/10, \qquad 0 \le t \le 1. \tag{4.6}$$

No convexity of the annulus is assumed. The potential is polynomial in real coordinates, so this segment argument needs no nonzero-phase condition at its interior points.

The onsite Hessian at $x_i \in \mathbb{R}^2$ is

$$\varepsilon\left[(|x_i|^2 - k_i)I_2 + 2x_i x_i^T\right], \tag{4.7}$$

and the graph Hessian is $-gL \otimes I_2$. For any real vector z_i, the onsite quadratic form is at most $\varepsilon(3|x_i|^2 - k_i)\|z_i\|^2$, by Cauchy–Schwarz and $\varepsilon > 0$. Since the largest eigenvalue of $-gL$ is $3g$, on every point of (4.6),

$$\nabla^2 \mathcal{V} \preceq \left[\frac{1}{10}\left(3\frac{121}{100} - \frac{999}{1000}\right) + \frac{3}{60}\right] I_6 = \frac{3131}{10000} I_6. \tag{4.8}$$

Only an upper Hessian bound is needed; no positive lower bound on the disk product is asserted. Taylor's integral formula and $b = -\nabla\mathcal{V}(\Omega)$ now give

$$\begin{aligned} \mathcal{V}(v) - \mathcal{V}(\Omega) &= -\|b\|_2^2 + \int_0^1 (1-t) b^T \nabla^2 \mathcal{V}(\Omega + tb) b \, dt \\ &\le -\left(1 - \frac{1}{2}\frac{3131}{10000}\right)\|b\|_2^2 = -\frac{16869}{20000}\|b\|_2^2. \end{aligned} \tag{4.9}$$

Synchronization and the equality states. The prestage is nonzero by (4.3), its phase width is $D_v \le D_0 < \pi/6$, and $\lambda = 1/300$ lies in the phase lemma's interval. The factor in (4.4) obeys

$$0 \le 1 - 9\lambda \cos 3D_v \le 1 - \frac{3\sqrt{2}}{200} < 1.$$

For the fixed prestage moduli $\rho_i = |v_i|$,

$$|\rho_i e^{i\psi_i} - \rho_j e^{i\psi_j}|^2 = (\rho_i - \rho_j)^2 + 2\rho_i\rho_j[1 - \cos(\psi_i - \psi_j)]. \tag{4.10}$$

The function $1 - \cos t$ increases with $|t|$ on this interval, so each pair distance is nonincreasing under the phase stage. Each onsite term is unchanged, and $g > 0$. Hence

$$\mathcal{V}(F\Omega) \le \mathcal{V}(v). \tag{4.11}$$

Combining with (4.9) proves (4.5), including the domain's closed boundaries. This uses the native first-harmonic graph distances, not merely the separate harmonic-three phase scalar.

If $\mathcal{V}(F\Omega) = \mathcal{V}(\Omega)$, the positive coefficient in (4.5) forces $b = 0$. Write $\Omega_i = r_i e^{i\theta_i}$ in a consistent interval and choose a channel m of minimal phase. The imaginary part of $e^{-i\theta_m} b_m = 0$ is

$$0 = g \sum_{j \ne m} r_j \sin(\theta_j - \theta_m). \tag{4.12}$$

Every summand is nonnegative, and any strictly larger phase makes its summand positive because the difference is at most D_0. Thus all phases agree. The remaining real equations are (3.1) in the amplitude cube, whose unique solution is the inherited r_*. Conversely, $e^{i\theta} r_*$ has $v = \Omega$ and zero kicks. Therefore

$$\mathcal{V}(F\Omega) = \mathcal{V}(\Omega) \iff b = 0 \iff \Omega \in \{e^{i\theta} r_*\}, \qquad \Omega \in \mathcal{D}. \tag{4.13}$$

□

The coefficient in (4.5) is neither a percentage reduction of potential nor a bound using $\|F\Omega - \Omega\|^2$. The result concerns the ideal mathematical map, not a certification of floating-point evaluation. It does not classify equality in every intermediate quantitative estimate.

5 Closed coherence coordinates

An autonomous retained description assigns the same next retained data to any two states with the same current retained data. This is the fibre criterion of [1], `thm:closure`. Full complex coherence and the real Gram already satisfy it on their regular domains. The question here is whether fewer existing raw pair entries suffice throughout the particular invariant domain $\mathcal{D}$.

Define

$$s_i = |\Omega_i|^2, \qquad C_{ij} = \overline{\Omega_i}\Omega_j = S_{ij} + iA_{ij}. \tag{5.1}$$

Choose channel 1 as a coordinate anchor and retain

$$y = \mathcal{R}(\Omega) = (s_1, s_2, s_3, x, y_a), \qquad x = A_{12}, \quad y_a = A_{13}. \tag{5.2}$$

The subscript distinguishes the second area from the tuple y. The anchor is bookkeeping, not a physical attachment or a symmetry assumption about the fixed unequal k.

Theorem 5.1 (Reconstruction and common-phase fibres)**.** *On $\mathcal{D}$, the tuple (5.2) determines full rank-one Hermitian coherence. With $p = S_{12}$ and $q = S_{13}$,*

$$p = +\sqrt{s_1 s_2 - x^2}, \qquad q = +\sqrt{s_1 s_3 - y_a^2}, \tag{5.3}$$

$$S_{23} = \frac{pq + xy_a}{s_1}, \qquad A_{23} = \frac{py_a - xq}{s_1}. \tag{5.4}$$

Its fibres on $\mathcal{D}$ are exactly common-phase orbits. These statements include all closed boundaries and zero-area cases.

Proof. Write $\Omega_i = r_i e^{i\theta_i}$ in an interval of width at most D_0. Every cosine of a relative phase is positive, so

$$S_{ij} = r_i r_j \cos(\theta_j - \theta_i) > 0, \qquad A_{ij} = r_i r_j \sin(\theta_j - \theta_i).$$

This fixes the positive roots in (5.3). Their radicands satisfy

$$s_1 s_2 - x^2,\ s_1 s_3 - y_a^2 \ \geq \left(\frac{81}{100}\right)^2 \cos^2 D_0 > 0. \tag{5.5}$$

Furthermore,

$$C_{23} = \frac{\overline{C_{12}} C_{13}}{s_1} = \frac{(p - ix)(q + iy_a)}{s_1},$$

which gives both signs in (5.4). The denominator $s_1 \geq 81/100$ never vanishes.

The source's matrix convention is

$$\mathcal{H} = \Omega\Omega^\dagger = S_G - iA_G, \qquad \mathcal{H}_{ij} = \overline{C_{ij}}, \qquad A_{ij} = -\operatorname{Im}\mathcal{H}_{ij}. \tag{5.6}$$

Thus the reconstruction is

$$\mathcal{H}(y) = \begin{pmatrix} s_1 & p - ix & q - iy_a \\ p + ix & s_2 & S_{23} - iA_{23} \\ q + iy_a & S_{23} + iA_{23} & s_3 \end{pmatrix}. \tag{5.7}$$

A convenient representative is

$$z(y) = \left(\sqrt{s_1}, \frac{p + ix}{\sqrt{s_1}}, \frac{q + iy_a}{\sqrt{s_1}}\right)^T. \tag{5.8}$$

It has squared moduli s_i, and $z(y)z(y)^\dagger = \mathcal{H}(y)$, proving rank-one positive semidefiniteness. For an original state, $z(\mathcal{R}(\Omega)) = e^{-i\theta_1}\Omega$. Equal retained tuples therefore give states differing by a common phase. Conversely every common rotation preserves the tuple and $\mathcal{D}$, proving the fibre statement. □

No area or phase difference is a denominator. Equal amplitudes and coincident phases do not cause singularities. Within this sector, $x = 0$ forces $\theta_2 = \theta_1$, and $y_a = 0$ forces $\theta_3 = \theta_1$; either can vanish without the other. If both vanish, all phases align and all reconstructed real coherences are the positive products $r_i r_j$. A rank-one real Gram means that the nonzero planar channel vectors are collinear; the narrow sector then forces the same ray, exactly this aligned case. The complex coherence is rank one everywhere, also at nonaligned states. Formula (5.8) recovers only a representative, not the discarded native common phase or its accumulated advance.

5.1 The exact image, rather than separate area bounds

Define first $\mathcal{Y} = \mathcal{R}(\mathcal{D})$.

Proposition 5.2 (Retained-domain characterization)**.** *A real tuple $y = (s_1, s_2, s_3, x, y_a)$ belongs to $\mathcal{Y}$ exactly when*

$$\frac{81}{100} \le s_i \le \frac{121}{100}, \tag{5.9}$$

the principal arcsines

$$\alpha_2 = \arcsin \frac{x}{\sqrt{s_1 s_2}}, \qquad \alpha_3 = \arcsin \frac{y_a}{\sqrt{s_1 s_3}} \tag{5.10}$$

are real, and

$$\boxed{\max(0, \alpha_2, \alpha_3) - \min(0, \alpha_2, \alpha_3) \le D_0.} \tag{5.11}$$

Proof. For a state in $\mathcal{D}$, subtract its channel-1 phase from the consistent lifts. Relative phases lie in $[-D_0, D_0]$, where principal arcsine inverts sine. The numbers in (5.10) are exactly those relative phases, proving necessity.

Conversely, under the stated conditions

$$(\sqrt{s_1}, \sqrt{s_2} e^{i\alpha_2}, \sqrt{s_3} e^{i\alpha_3})$$

belongs to $\mathcal{D}$ and has retained areas x, y_a. Its positive cosines identify it with (5.8), proving sufficiency. In particular, every tuple in this characterized image has the valid coherence (5.7). □

Real arcsines require $|x| \le \sqrt{s_1 s_2}$ and $|y_a| \le \sqrt{s_1 s_3}$; the span condition strengthens both normalized bounds to $\sin D_0$. Even these separate stronger bounds would be insufficient: normalized areas $\sin D_0$ and $-\sin D_0$ recover phases spanning $2D_0$. The joint condition (5.11) excludes this unintended enlargement.

5.2 The induced map and repeated regularity

Theorem 5.3 (Autonomous retained evolution)**.** *For reconstructed $\mathcal{H}(y)$, define*

$$\begin{aligned} M(y) &= I + \varepsilon \operatorname{diag}(k - s) + gL, \\ \widetilde{\mathcal{H}} &= M \mathcal{H} M^T, \qquad \rho_i = \sqrt{\widetilde{\mathcal{H}}_{ii}} > 0, \\ \delta_i &= \lambda \sum_{j \ne i} \operatorname{Im} \left[\left(\frac{\widetilde{\mathcal{H}}_{ji}}{\rho_j \rho_i} \right)^3 \right], \qquad U_\delta = \operatorname{diag}(e^{i\delta_i}), \end{aligned} \tag{5.12}$$

$$\mathcal{H}^+ = U_\delta \widetilde{\mathcal{H}} U_\delta^\dagger. \tag{5.13}$$

Then

$$F_{\mathcal{R}}(y) = (\mathcal{H}^+_{11}, \mathcal{H}^+_{22}, \mathcal{H}^+_{33}, -\operatorname{Im} \mathcal{H}^+_{12}, -\operatorname{Im} \mathcal{H}^+_{13}) \tag{5.14}$$

is well defined on $\mathcal{Y}$, with

$$\mathcal{R}(F\Omega) = F_{\mathcal{R}}(\mathcal{R}(\Omega)) \qquad (\Omega \in \mathcal{D}),$$
$$F_{\mathcal{R}}(\mathcal{Y}) \subseteq \mathcal{Y}. \tag{5.15}$$

All subsequent mathematical iterates are justified on that image.

Proof. For any representative $\Omega \in \mathcal{D}$, $M\Omega = v$ and $\widetilde{\mathcal{H}} = vv^{\dagger}$. Thus $\rho_i = |v_i| > 0$ by the actual inherited prestage bounds (4.3). The transposed entry gives

$$\frac{\widetilde{\mathcal{H}}_{ji}}{\rho_j \rho_i} = \frac{v_j \overline{v}_i}{|v_j||v_i|} = e^{i(\psi_j - \psi_i)}.$$

Cubing and taking its imaginary part produces the native kick with the correct sign. Equations (5.12)–(5.13) are precisely the regular coherence update of [1], `thm:Hclosure`. Extraction using (5.6) proves the first part of (5.15). Equivalently, equal tuples are common-phase orbits and the regular map is equivariant, so the fibre criterion proves representative independence.

For $y \in \mathcal{Y}$, choose $\Omega \in \mathcal{D}$ with $\mathcal{R}(\Omega) = y$. Invariance of $\mathcal{D}$ gives $F\Omega \in \mathcal{D}$, hence $F_{\mathcal{R}} y \in \mathcal{Y}$. Repeating this argument proves

$$F_{\mathcal{R}}^{n}(\mathcal{R}(\Omega)) = \mathcal{R}(F^{n}\Omega), \qquad n \geq 0, \tag{5.16}$$

with positive prestages at every step. This is an induction, not a computed trajectory. □

The ji index and the minus signs in (5.14) must be preserved. Replacing $\mathcal{H}_{ij}$ by C_{ij} would reverse the phase reconstruction. The native zero convention is untouched: this proof avoids zero prestages by a proved invariant domain, not by changing the map. The lambda-zero bypass is consistent with the identity phase stage, although the present tuple has $\lambda = 1/300$.

6 Retained energies and transfer of descent

The coordinate construction retains more than an autonomous update. It preserves the scalar that motivated the reduction and every individual graph contribution, rather than only their sum.

Theorem 6.1 (Exact energy retention)**.** *For every $y \in \mathcal{Y}$, define*

$$E_{ij}(y) = s_i + s_j - 2S_{ij}(y), \tag{6.1}$$

$$\widehat{\mathcal{V}}(y) = \varepsilon \sum_i \left(\frac{s_i^2}{4} - \frac{k_i s_i}{2} \right) + \frac{g}{2} \sum_{i<j} E_{ij}(y), \tag{6.2}$$

$$N(y) = M(y) - I, \qquad \widehat{\mathcal{B}}(y) = \operatorname{tr}(N\mathcal{H}N^{T}). \tag{6.3}$$

For any representative $\Omega \in \mathcal{D}$, these satisfy

$$E_{ij}(y) = |\Omega_i - \Omega_j|^2 = \|D_i\Omega_i - T_{ij}D_j\Omega_j\|^2, \tag{6.4}$$

$$\widehat{\mathcal{V}}(\mathcal{R}(\Omega)) = \mathcal{V}(\Omega),$$

$$\widehat{\mathcal{B}}(y) = \operatorname{tr}(NS_G N^{T}) = \|v - \Omega\|_2^2 \geq 0. \tag{6.5}$$

Here D_i and T_{ij} are the accepted matched decoder and transport.

Proof. Expansion of a squared complex distance proves (6.1). In the adopted tangent frames of [1], `eq:decoder` and `eq:faceupdate`, D_i is a real-linear isometry and $T_{ij}D_j = D_i$. Thus the transported difference is $D_i(\Omega_i - \Omega_j)$, proving (6.4). Substitution into the edge-once potential gives (6.2).

Because N is real and $N\Omega = v - \Omega$,

$$N\mathcal{H}N^T = (N\Omega)(N\Omega)^\dagger.$$

This matrix is Hermitian positive semidefinite, and its trace is the squared complex Euclidean norm. Moreover, $A_G^T = -A_G$, so NA_GN^T is real skew-symmetric with zero diagonal. Its trace vanishes, proving the equality with $\mathrm{tr}(NS_GN^T)$. No alignment assumption is used, and symmetry of N is unnecessary for this argument. □

The retained matched distances do not reconstruct the common-phase-dependent face vectors themselves. Their identification with transported tangent differences supplies no physical flux law.

Corollary 6.2 (Descent on the retained image)**.** *For every $y \in \mathcal{Y}$,*

$$\boxed{\widehat{\mathcal{V}}(F_{\mathcal{R}}y) - \widehat{\mathcal{V}}(y) \le -\frac{16869}{20000}\widehat{\mathcal{B}}(y).} \tag{6.6}$$

Let

$$y_* = (r_{*,1}^2, r_{*,2}^2, r_{*,3}^2, 0, 0). \tag{6.7}$$

Then

$$\widehat{\mathcal{V}}(F_{\mathcal{R}}y) = \widehat{\mathcal{V}}(y) \quad \Longleftrightarrow \quad y = y_*. \tag{6.8}$$

The point y_* is the unique fixed retained state.

Proof. Choose any $\Omega \in \mathcal{D}$ representing y. Semiconjugacy and Theorem 6.1 transfer Theorem 4.1 directly to (6.6). The equality circle in (4.13) maps to the single tuple (6.7); conversely that tuple reconstructs $r_*r_*^T$, whose fibre is exactly the circle. This proves (6.8). Its image is fixed; any other fixed retained state would also have equal potential, a contradiction. □

Likewise $\widehat{\mathcal{B}} = 0$ exactly at y_*, by (6.5) and (4.13). No root solve, new decay-rate estimate, or classification of equality in every quantitative bound is needed for this transfer.

7 Attained minimum raw-entry counts

All three intensities are now compulsory, and only the number of additional entries selected from

$$\mathcal{E} = \{S_{12}, S_{13}, S_{23}, A_{12}, A_{13}, A_{23}\} \tag{7.1}$$

is minimized. A complex pair entry counts as two real entries. A selected tuple must have an autonomous update on its actual image of $\mathcal{D}$, determine the current potential and all three individual edge energies, and determine $\|v - \Omega\|_2^2$ so that the accepted descent and equality statements are expressible from retained data alone.

For a subdictionary $\mathcal{A}$, let $m(\mathcal{A})$ be the smallest cardinality of a subset satisfying this joint objective throughout $\mathcal{D}$. This is not minimization over arbitrary encodings, histories, clocks, observer states, or special strata. A tuple can be excluded by failing any one requirement.

Theorem 7.1 (Minimum with signed entries allowed)**.** *For the fixed coefficients and entire domain of Section 4,*

$$m(\mathcal{E}) = 2,$$

attained by A_{12}, A_{13} together with the three intensities.

Proof. Theorems 5.1, 5.3, and 6.1, with Corollary 6.2, establish sufficiency.

For the lower bound, let a proposed singleton concern pair $\{i, j\}$, and let ℓ be the remaining channel. Fix arbitrary labelled amplitudes $r_a \in [9/10, 11/10]$ and keep all labelled coefficients k_a and all other parameters unchanged. Consider the static family

$$\Omega_i(t) = r_i, \qquad \Omega_j(t) = r_j, \qquad \Omega_\ell(t) = r_\ell e^{it}, \qquad 0 \le t \le D_0. \tag{7.2}$$

Every state belongs to $\mathcal{D}$. Intensities are constant, and both possible types of entry on the retained pair are constant:

$$S_{ij}(t) = r_i r_j, \qquad A_{ij}(t) = 0.$$

The other edge energies obey

$$\begin{aligned} E_{i\ell}(t) &= (r_i - r_\ell)^2 + 2r_i r_\ell(1 - \cos t), \\ E_{j\ell}(t) &= (r_j - r_\ell)^2 + 2r_j r_\ell(1 - \cos t). \end{aligned} \tag{7.3}$$

The onsite potential and the ij edge are unchanged, hence

$$\mathcal{V}(\Omega(t)) - \mathcal{V}(\Omega(0)) = g r_\ell (r_i + r_j)(1 - \cos t) > 0 \quad (0 < t \le D_0). \tag{7.4}$$

This excludes any of the six singleton choices by current-energy retention. It also excludes retaining only intensities. Relabelling the displayed static family does not permute k or assert a permutation symmetry of the dynamics. □

The aligned retained pair in (7.2) is a valid part of the required domain, not a removable generic-chart exception. The lower bound does not claim that each special stratum separately needs two entries.

7.1 Real coherences alone

Let

$$\mathcal{E}_{\text{real}} = \{S_{12}, S_{13}, S_{23}\}.$$

The objective is unchanged; recovering signed complex coherence is *not* an extra requirement in this comparison.

Theorem 7.2 (Real-only minimum)**.** *On the same domain and at the same coefficients,*

$$m(\mathcal{E}_{\text{real}}) = 3.$$

The full real Gram attains this minimum, although it discards signed orientation.

Proof. All three real coherences and the intensities determine S_G. The inherited real-Gram closure theorem [1], `thm:Sclosure`, applies wherever $(MS_GM^T)_{ii} > 0$, including rank-deficient factors. On the actual image of $\mathcal{D}$, the bounds (4.3) and invariance supply persistent regularity and an invariant image. Equal Gram factors differ by an $O(2)$ transformation, meaning common phase and/or conjugation; both preserve $\mathcal{D}$ and have the required regular-map equivariance. Thus the update is autonomous on that image.

All three edge energies and the potential are read directly from S_G, and the prestage norm is $\operatorname{tr}(NS_GN^T)$ by (6.5). The same descent and equality statement transfers, with the fixed circle represented by $r_*r_*^T$.

To exclude two real entries, note that any two triad edges share one channel h. Let the other labels be j, ℓ. Fix amplitudes as before and choose $u, v > 0$ with $u + v \le D_0$. Compare the static states

$$\Omega_h^{\pm} = r_h, \qquad \Omega_j^{\pm} = r_j e^{iu}, \qquad \Omega_\ell^{\pm} = r_\ell e^{\pm iv}. \tag{7.5}$$

The plus state's width is $\max(u, v)$, and the minus state's width is $u + v$; both are in $\mathcal{D}$. Their intensities agree, as do the retained entries:

$$S_{hj}^{\pm} = r_h r_j \cos u, \qquad S_{h\ell}^{\pm} = r_h r_\ell \cos v.$$

The omitted entry differs:

$$\begin{aligned} S_{j\ell}^{+} &= r_j r_\ell \cos(u - v), & S_{j\ell}^{-} &= r_j r_\ell \cos(u + v), \\ S_{j\ell}^{+} - S_{j\ell}^{-} &= 2 r_j r_\ell \sin u \sin v > 0. \end{aligned} \tag{7.6}$$

Therefore

$$\begin{aligned} E_{j\ell}^{-} - E_{j\ell}^{+} &= 4 r_j r_\ell \sin u \sin v > 0, \\ \mathcal{V}(\Omega^{-}) - \mathcal{V}(\Omega^{+}) &= 2 g r_j r_\ell \sin u \sin v > 0. \end{aligned} \tag{7.7}$$

All other edge and onsite contributions agree. The labelled cases are

h	(j, ℓ)	retained entries
1	$(2, 3)$	S_{12}, S_{13}
2	$(1, 3)$	S_{12}, S_{23}
3	$(1, 2)$	S_{13}, S_{23}

and every case keeps the original unequal k fixed. Counts zero and one are already excluded by Theorem 7.1. □

Table 2: Comparison for the joint objective, always retaining all three intensities. Failed rows assert energy-retention failure, not dynamic nonclosure.

Additional entries	Autonomous and energy-retaining status	Count conclusion
None	Same intensities, different current potential in (7.2)	Insufficient
Any single entry	Same retained pair, different current potential in (7.4)	Insufficient
A_{12}, A_{13}	Full coherence reconstruction, invariant update, all required energies	Attains minimum 2 in $\mathcal{E}$
Any two real entries	Omitted edge energy and potential differ in (7.7)	Insufficient
All three real entries	Inherited real-Gram closure; all required energies	Attains minimum 3 in $\mathcal{E}_{\text{real}}$

The mechanism is elementary but domain-dependent. Two anchor cosines do not tell on which side of the anchor each other phase lies; changing one side alters the missing relative cosine. Signed sines retain that side information, and the narrow interval makes them invertible. The signed candidate can therefore use fewer raw entries while retaining orientation that the real Gram discards. It is not thereby the coarsest sufficient energy quotient. No classification of all mixed two-entry subsets is required for either attained minimum.

8 Interpretation and limits

The main connection is now explicit: the existing potential controls the complete response in two stated settings, and a smaller collection of existing pair data preserves both autonomous evolution and that control on the finite domain.

The local result depends on the particular triad quotient and the actual potential Hessian. It does not assert descent for all stable matrices, larger rings, other branches, or unit-modulus boundaries. The finite result uses a specified coefficient tuple and the entire closed 15-degree coherent domain. Its phase-sector hypothesis cannot be replaced by arbitrary phase dispersion. That domain is distinct from the 60-degree finite-return sector noted in [3, Section 8] and from any spatial cone angle.

The two minimum statements compare raw-entry subsets only. They do not optimize over arbitrary nonlinear encodings or establish a universally minimal observable. Their exclusion proofs compare current energies of static states; no next-state discrepancy is inferred for every rejected tuple. The full real Gram is already a sufficient energy-retaining quotient despite losing signed orientation.

The native potential, intensity, and transported squared differences are distinct mathematical quantities. Their identities do not supply physical mass, energy, flux, pose, or a gravitational field law. The adopted decoder and existing observer constructions already describe their specified native and readout evolution; an unresolved physical correspondence does not erase that evolution. The channel anchor remains a coordinate choice, and the supporting 180-degree line remains stationary.

The representative $z(y)$ does not recover the original common phase or its accumulated advance. That reconstruction question, spatial and fixed-support correspondence, outer geometry, and legitimate SRG integration remain separate later questions. Gravity–magnetism coupling remains a linked programme objective; the intelligence, memory, games, and embodiment programme remains parallel. No new interpretation or successor investigation is needed to state the present theorems.

The archived symbolic helpers support identities used in the accepted proof notes. They are not substitutes for the continuous-domain proofs and were not rerun for this manuscript. Normal document compilation and layout checks do not constitute scientific certification of floating-point trajectories.

A Finite-step pair-coherence identity

The following algebra fixes the sign and normalization independently of any descent assertion.

Lemma A.1. *With tildes denoting prestage quantities and $d_{ij} = \delta_j - \delta_i$,*

$$S_{ij}^{+} = \widetilde{S}_{ij} \cos d_{ij} - \widetilde{A}_{ij} \sin d_{ij}, \tag{A.1}$$

$$\mathcal{V}(F\Omega) - \mathcal{V}(v) = -g \sum_{i<j} \left[\widetilde{S}_{ij}(\cos d_{ij} - 1) - \widetilde{A}_{ij} \sin d_{ij} \right]. \tag{A.2}$$

Proof. By the convention (5.1),

$$\overline{F_i} F_j = \overline{v}_i v_j e^{i(\delta_j - \delta_i)} = (\widetilde{S}_{ij} + i\widetilde{A}_{ij})(\cos d_{ij} + i \sin d_{ij}).$$

Taking real parts proves (A.1). Synchronization preserves each intensity. Since $|z_i - z_j|^2 = |z_i|^2 + |z_j|^2 - 2S_{ij}$, the edge-once coefficient $g/2$ contributes $-g$ times the coherence change, proving (A.2). □

The identity alone has no global sign. The local Hessian argument and the finite-domain pair-angle lemma supply their respective signs under different hypotheses.

B Proof ownership and reproducibility

The admitted branch, weighted prestage response, exact positive-triad spectrum, and diagonal-cycle criterion are inherited from [1], respectively `thm:branch`, `eq:weighted`, `thm:triad`, and `thm:cycle`. The same work supplies matched transport and regular closure through `eq:decoder`, `eq:faceupdate`, `thm:Hclosure`, and `thm:Sclosure`. The invariant-domain and profile inputs come from [2], Sections 6–7. Their hypotheses are stated at the points of use.

The local and finite potential deductions and Appendix A were accepted in the first proof note [3]; reconstruction, retention, and the minimum-count statements were accepted in the second [4]. Both notes and their original helper/result pairs are included in distinct evidence directories. One editorial comma in the second note's iterate exponent was deleted before copying the corrected note; the original bytes and exact one-byte receipt are preserved in the review package. No mathematical content changed.

The package's claim/source ledger maps the manuscript labels to the accepted proof-note equations and inherited source labels. Its preparation report records input identities, compiler and rendering details, and the final page inspection. These process records are separate from the proofs. The paper's source compiles independently of the scientific helpers; those archival files are not document-build dependencies.

References

[1] Tri-Octagon research programme. *Native Response Geometry and Spectral Circulation in the Tri-Octagon Map*. Research publication v0.1, 6 October 2026. Accepted reading edition `82cab10cbe55`. Response chapter; spectral chapter; transport and closure chapter.

[2] H. F. Halldórsson. *Transient Collective Turning and Coherent Relaxation in the Native Triad*. Repository research edition v0.1, 28 October 2026. Sections 6–7, accepted reading edition `7cc1fa03af58`. Manuscript source.

[3] Tri-Octagon research programme. *Native potential descent: independent proof review*. Task `NATIVE_ENERGY_RESPONSE_01`, 1 November 2026. Codex proof verification under GPT scientific direction; project-internal mathematical acceptance recorded in the preparation order. Packaged proof note. Unpublished investigation record.

[4] Tri-Octagon research programme. *Minimum closed native information: independent proof review*. Task `NATIVE_CLOSED_INFORMATION_01`, 2 November 2026. Codex proof verification under GPT scientific direction; project-internal mathematical acceptance recorded in the preparation order. Packaged proof note, with the authorized single-comma correction. Unpublished investigation record.